

Capstone: Proposal

Topic

The economics of car dependency in America and how oil, institutions, industry and infrastructure led to a system where the average American finds transportation as unavoidable, expensive and unequal. My capstone project will follow how this system was implemented into our society and how it shows up in modern society. Then, it splits into two futures: one dystopian one utopian

Lens Used

- **Challenging Mental Models:** Challenge and debate the normalization of car ownership in America. Shows how car ownership leads to freedom by showcasing how it becomes an economic staple that was implemented and infrastructured
- **Data Visualization:** Timelines, mapping data/systems (Statistics, costs, oil)
- **Utopian/Dystopian:**
 - Dystopia: Car dependency rises therefore also increases inequality. Ownership becomes harder for many
 - Utopia: walkable cities, mixed neighborhoods, free transit

Core Narrative Idea

Car dependency in America did not just happen naturally, it was something that was designed and implemented. Some important factors include:

- Creation/Expansion of the interstate highway system
- Creation/Designing the suburban lifestyle
- US Economy dependence on oil
- Rise of auto industry (Influence/Economic Demand and Market)

- Zoning laws such as redlining that designate living/working/commerce areas

Timeline

Past/Historical: System formation and Design

- Post WWII highway expansion
- Implementation of zoning laws and growth of suburban neighborhoods
- Destruction of public transportation
- Oil Dependency rise (economically/society)
- Rise of the Auto Industry (Culture/Policy)

Present: Hidden Costs

- Car loans, insurance, maintenance, gas are unavoidable/take a huge chunk of income
- Factor in commuting times/commuting as unpaid labor
- Fluctuation of oil prices affecting everyday living costs + American economy relies on the oil industry
- Rise of inequality: access to mobility increases opportunities, no mobility = decreased opportunities

Future Split

- Dystopia:
 - car subscription based/rise of rent economy
 - Cars > people, but less and less car owners
 - Public space optimized for car usage/tracking
 - Huge oil scarcity, costs worsen the American economy
 - Mobility becomes a privilege
- Utopia
 - Major investments in Public transportation
 - Walkable and dense cities, car ownership is an option
 - Reduction of oil usage and scarcity

- Highways reclaimed by public spaces/housing/parks

Interaction Ideas: Updated Timelines

- **Scroll Based:** Past > Presents > User chooses to split between keeping/discarding car (Keep: Dystopia, Discard: Utopia)
- **Different Versions of Timeline:** One that focuses on policy implementations vs oil shocks vs historical events,,
- **Cost Calculator:** Side bar that as you scroll, shows you the amount of money that has been put towards cars/oils vs public transportation
- **Hidden Vs Visible Costs:** Toggle that allows users to switch between hidden (emotional, land, time) and visible costs
- **Overall:** Users input or explore transportation scenarios to understand personal and systemic costs of car dependency and compare alternative futures.

Possible JS/Tools:

- **GSAP/Scroll Trigger:** Scrolling Transitions
- **P5.js:** visualization of data (car density, oil flows, etc)
- **D3.js:** timelines, economic data visualization

Target Audience:

- Broad: People living in the US
- Narrow: Young Adults/Students making the decisions about owning a car
- Very Specific: College urban/suburban students who assume car ownership is “normal” and unavoidable

Key Tensions: Car dependency often feels like a choice that we make, when in fact is a system that was designed and decided for us.